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A selective gate-recess and passivation integrated process for high-performance enhancement-mode GaN HEMTs on an ultrathin barrier

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ABSTRACT

Enhancement-mode (E-mode) operation is essential for GaN high-electron-mobility transistors (HEMTs) to enable power-efficient and simplified RF circuit design. Conventional gate-recess techniques often face trade-offs among threshold voltage uniformity, channel damage, and access region resistance, especially when employing ultrathin barriers for scaling. In this work, we present an integrated fabrication process combining an ultrathin InAlN/AlN composite barrier, a highly selective gate recess etch, and a plasma-enhanced chemical vapor deposition silicon nitride passivation layer. The highly selective etch (GaN/InAlN selectivity >19.5) enables precise removal of the GaN cap with minimal channel damage, while the passivation layer effectively enhances the two-dimensional electron gas conductivity in the access regions. The fabricated E-mode HEMTs with a 130-nm gate demonstrate a threshold voltage of +0.25 V, a maximum saturation drain current density of 560 mA/mm, and excellent RF performance with current gain cut-off/maximum oscillation frequencies (f_T/f_{max}) of 85/235 GHz for the 130-nm gate device. The demonstrated performance underscores the effectiveness of this integrated process and its potential for enabling high-frequency, enhancement-mode GaN electronics.

1. Introduction

Over the past decade, gallium nitride (GaN)-based high electron mobility transistors (HEMTs) have been demonstrated to exhibit excellent radio-frequency (RF) power switching and high-frequency performance, owing to their wide bandgap, high critical electric field, and inherent polarization effects [1,2]. This polarization effect, which originates from the polar properties of III-N materials, presents a significant challenge for the fabrication of enhancement-mode (E-mode) GaN HEMTs [3,4]. Consequently, the majority of reported devices to date operate in depletion-mode (D-mode) [1,5–7]. However, E-mode HEMTs are of significant importance for expanding the applications of HEMTs. The circuit design can be simplified by using E-mode devices, as passive components typically required for biasing in high-frequency circuits often introduce substantial losses [4,8]. Furthermore, E-mode HEMTs are highly favored for power switching applications due to their inherent fail-safe operation and low power consumption [9]. Recently, Then et al. demonstrated a GaN chiplet technology on 300 mm GaN-on-Si with comprehensive reliability studies, further advancing E-mode GaN HEMTs for high-frequency and power applications [10].

Among the various techniques for achieving E-mode operation, such as p-GaN gate and fluorine implantation, the recessed-gate approach is particularly favored for RF applications due to its compatibility with high-frequency performance [11–19]. This method relies on thinning the gate region barrier layer to deplete the two-dimensional electron gas (2DEG). However, conventional recess etching is typically time-controlled, leading to sensitivity to process variations. This often results in non-uniform threshold voltage (V_{th}) and transconductance (g_m), and often introduces plasma-induced damage that degrades channel mobility and ultimate frequency performance [20–23].

Highly selective etching processes that automatically terminate at the barrier/cap interface have been developed to address these issues. For instance, inductively coupled plasma (ICP) etching with high GaN/InAlN selectivity enables self-termination, providing precise barrier thickness control and minimal channel damage [24–26]. This technique facilitates the use of ultrathin barriers, which are intrinsically depleted

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and thus naturally suited for E-mode operation. However, a fundamental drawback of such ultrathin-barrier heterostructures is the severely limited 2DEG conductivity in the source/drain access regions, leading to high parasitic resistance [3,27].

Effective surface passivation is therefore essential to recover access region conductivity. Studies have shown that dielectric layers such as silicon nitride (SiN) can significantly enhance 2DEG transport properties in thin-barrier structures through polarization and interface charge effects [28–33]. For example, passivation has been shown to reduce sheet resistance (R_{sh}) by orders of magnitude and increase 2DEG density in ultrathin-barrier devices [3,27]. Therefore, the successful integration of a highly selective, self-terminating gate recess with an effective passivation scheme is a critical pathway to realizing high-performance E-mode HEMTs, yet a systematic demonstration combining these elements remains necessary.

In this work, we demonstrate this integrated approach by fabricating E-mode HEMTs based on an ultrathin-barrier heterostructure. The key fabrication steps consist of selectively removing the GaN cap layer to leave an ultrathin barrier (1 nm InAlN/1 nm AlN) under the gate using a highly repeatable etching process, and depositing a 20 nm PECVD SiN film that serves as both a passivation and protection layer, effectively enhancing 2DEG transport in the access regions. The resulting device demonstrates a threshold voltage of 0.25 V, a maximum saturation drain current density of 560 mA/mm, a current gain cut-off frequency (f_T) of 85 GHz, and a maximum oscillation frequency (f_{max}) of 235 GHz, which confirms its potential for high-speed mixed-signal GaN-based circuits.

2. Experimental

2.1. Epitaxial structure and material properties

The epitaxial structure was grown on a sapphire substrate via metal-organic chemical vapor deposition (MOCVD). The layer sequence consisted of a 1.5-μm semi-insulating GaN buffer, a 300-nm GaN channel, a composite barrier of 1-nm InAlN/1-nm AlN, and a 2-nm GaN cap layer. Hall measurements of the as-grown material revealed a 2DEG sheet density of $3.6 \times 10^{11}\ cm^{-2}$ and a high R_{sh} of $4.5 \times 10^4\ \Omega/\square$, consistent with the expected carrier depletion in the ultrathin-barrier access regions.

2.2. Device fabrication process

Device fabrication commenced with the deposition of a silicon dioxide (SiO_2) layer by plasma-enhanced chemical vapor deposition (PECVD). Electron-beam lithography (EBL) and reactive ion etching were used to define a SiO_2 hard mask, which patterned a 350-nm source-drain distance and regions for subsequent regrowth. A 50-nm silicon-doped n^+-GaN layer was selectively regrown by MOCVD at 930 °C to form low-resistance ohmic contact areas. The SiO_2 mask was then removed using a buffered oxide etch (BOE), followed by mesa isolation via ICP etching. Ohmic contacts were formed by electron-beam evaporation of Ti/Au (20/200 nm) and liftoff.

2.3. Gate formation with selective recess etching

A T-shaped gate with a foot length (L_g) of 130 nm and a total width of 2 × 20 μm was defined by EBL. The key gate recess was formed using a highly selective ICP etching process with SF_6-based chemistry. This process is characterized by a high GaN-to-InAlN etch selectivity (>19.5:1), with GaN and InAlN etch rates of 10 nm/min and 0.51 nm/min, respectively (Fig. 1). This high selectivity enables the precise, self-terminating removal of the 2-nm GaN cap layer, thereby exposing the underlying ultrathin InAlN/AlN barrier with minimal damage to preserve channel integrity. Considering the ±0.5 nm growth tolerance of the GaN cap, a fixed etch time of 20 s was applied, which may result in a slight over-etch. Immediately following the recess, the Ni/Au (20/250 nm) gate metal was deposited by electron-beam evaporation and liftoff.

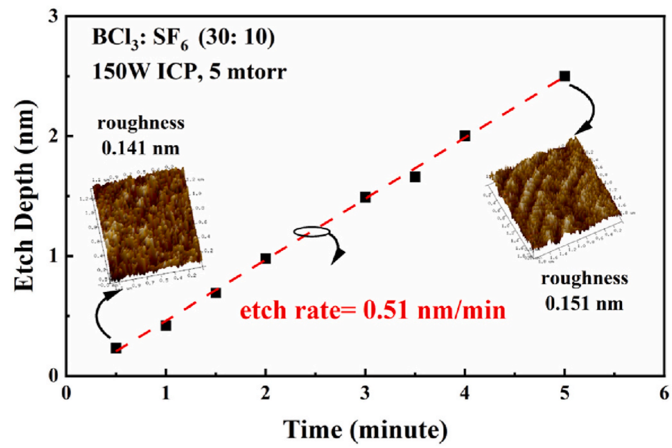


Fig. 1. Etch depth dependency over time for recessed InAlN, demonstrating high selectivity (>19.5:1).

2.4. Passivation and final processing

Finally, a 20-nm-thick PECVD SiN layer was deposited and the SiN in the pad region was removed by BOE. This layer serves the dual purpose of surface passivation and polarization enhancement. Hall measurements confirmed the effectiveness of the passivation, showing a significant increase in the 2DEG sheet density to $7.9 \times 10^{12}\ cm^{-2}$ and a corresponding reduction in sheet resistance to 401 $\Omega/\square$ in the access regions. A schematic of the completed device is shown in Fig. 2. The fabricated devices were then characterized for their DC, RF, and surface morphological properties, as detailed in the following section.

3. Results and discussion

3.1. Surface morphology and etch quality

The structural and compositional characteristics of the gate region are detailed in Fig. 3. Fig. 3(a) is a scanning electron microscopy (SEM) image of the T-gate, with the red rectangle marking the specific area from which the cross-sectional TEM sample was extracted. The cross-sectional transmission electron microscope (TEM) image in Fig. 3(b) shows the recessed area, where the measured ultrathin InAlN/AlN barrier (approximately 2.3 nm in thickness) remains intact with a sharp interface, confirming the self-terminated removal of the GaN cap. Energy-dispersive X-ray spectroscopy (EDS) elemental maps (Fig. 3(c, d)) under the gate provide further evidence: the Ga signal is depleted in the recess. While the Al map shows its strongest intensity localized within the barrier layer, a faint upward diffusion signal is also observed. Critically, there is no downward extension of the primary Al signal,

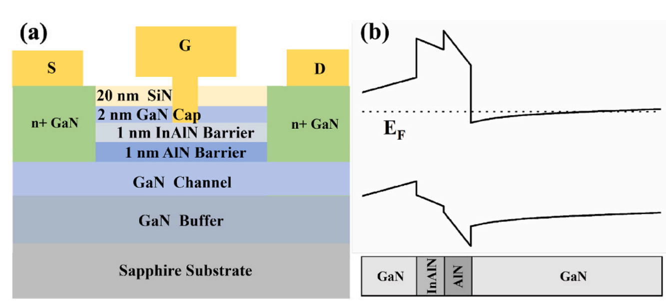


Fig. 2. (a) Schematic drawing of the E-mode HEMT and (b) energy band diagram.

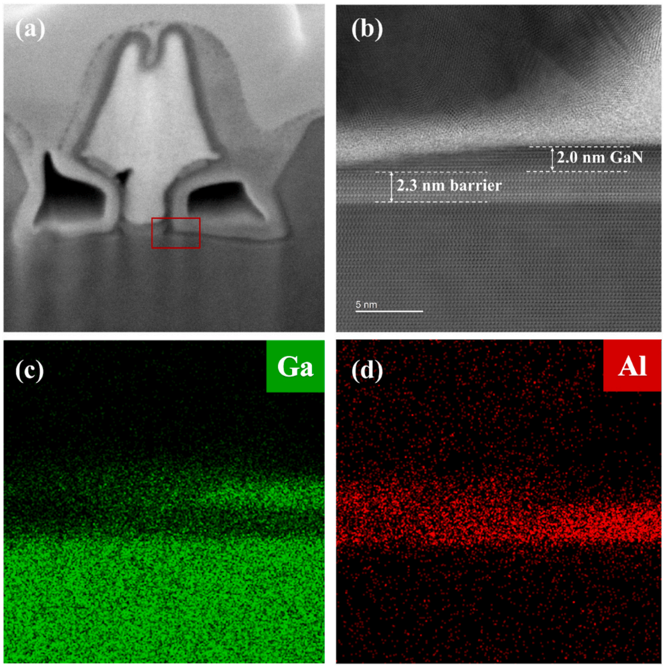


Fig. 3. Gate region characterization. (a) SEM image of the fabricated T-shaped gate. The red rectangle indicates the area where the cross-sectional TEM sample was lifted out. (b) Cross-sectional TEM image showing the GaN cap recess and the intact ultrathin InAlN/AlN barrier after metallization. Corresponding EDS elemental maps of the region under the gate for (c) Ga and (d) Al. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

confirming that the recess etch did not penetrate the barrier. The barrier's lattice fringes in the high-resolution TEM image are also well-resolved, indicating minimal crystallographic damage.

To evaluate the surface integrity after etching, atomic force microscopy (AFM) was performed on a representative area etched under identical conditions. The root-mean-square (RMS) roughness measured in this etched region is 0.269 nm, which is comparable to, and slightly lower than, the 0.284 nm RMS roughness of the pristine GaN cap surface. It should be noted that the RMS value reported in Fig. 1 (∼0.23 nm) was measured on a large-area test coupon, representing the intrinsic smoothness of the etching process, while the value of 0.269 nm reported here was measured on the actual sub-micrometer gate recess region (130 nm × 20 μm) on the device wafer. The slightly higher RMS in the device recess is attributed to microloading effects and mask-edge-induced local non-uniformities, which are inherent to small-feature etching. This maintained smoothness indicates that the selective etching process introduces minimal additional roughness, supporting its low-damage nature which is crucial for preserving channel mobility.

3.2. Electrical characteristics and contact resistance

To evaluate the impact of passivation on access resistance, transmission-line measurements (TLMs) were performed with gap spacings of 2, 6, 10, 15, and 20 μm, defined by the edges of the regrown n^+-GaN contacts (Fig. 4). The measurements yielded a R_{sh} of 430 Ω/□, and an access resistance (R_{ac}, from the ohmic metal to the gate edge) of 0.434 Ω mm. The extracted R_{sh} of 430 Ω/□ represents a significant reduction from the value of $\sim 10^4$ Ω/□ measured before passivation. This quantitative improvement, along with the low R_{ac}, directly confirms the effectiveness of the PECVD SiN layer in restoring 2DEG conductivity in the access regions, which is critical for achieving the low on-resistance (R_{on}) observed in the devices. The enhancement of 2DEG conductivity is primarily attributed to the electrostatic induction effect

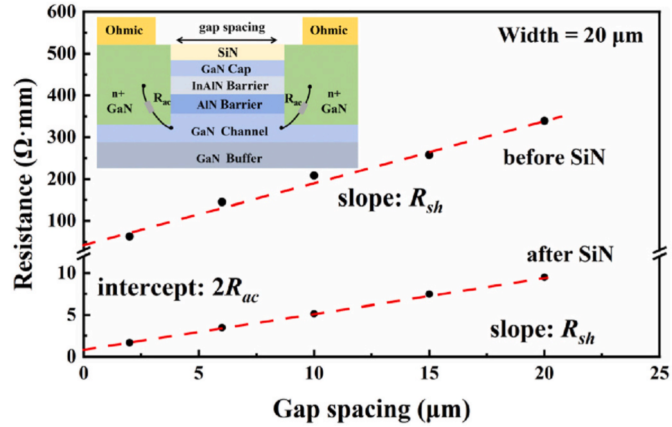


Fig. 4. TLMs of the E-mode HEMT structure before and after SiN passivation with a measurement schematic.

of the PECVD SiN layer. Fixed positive charges within the dielectric generate an additional vertical electric field at the interface, which further bends the energy bands of the underlying heterostructure and leads to an increased 2DEG concentration in the access regions [34,35]. The low access resistance achieved through passivation directly contributes to the favourable DC performance observed in the devices, as shown in Fig. 5 of following discussion.

The DC characteristics of the device were measured using a Keithley 4200A-SCS parameter analyzer. Fig. 5 presents the output and transfer characteristics of the device. An on-resistance of 2.0 Ω mm was extracted. A maximum output drain current density of 560 mA/mm was achieved at $V_{gs} = 2$ V and $V_{ds} = 5$ V, which is 40 times the value measured before passivation. The slight non-linearity at low drain bias is attributed to the moderate 2DEG density and non-ideal injection from the regrown n^+-GaN contact. According to the quantum contact resistance

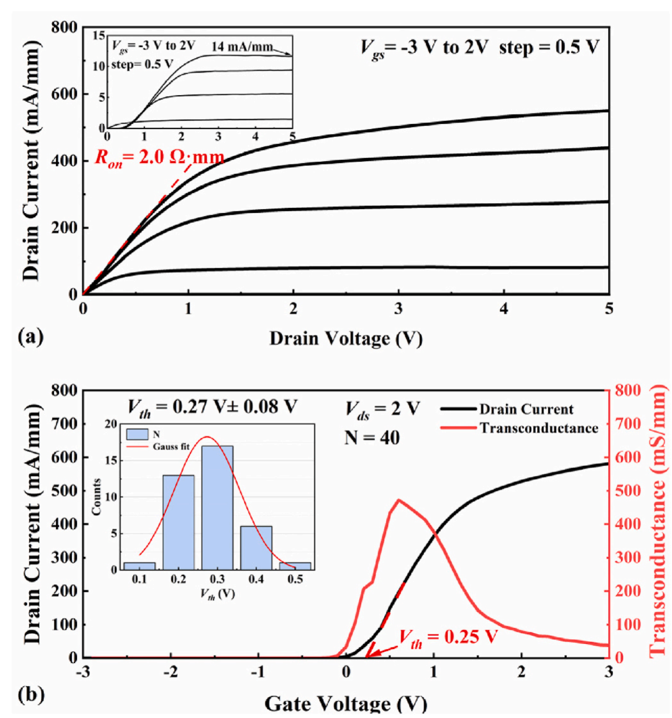


Fig. 5. (a) Output characteristics of the E-mode HEMT after SiN passivation, with the inset showing the results before SiN passivation, and (b) transfer characteristics of the E-mode HEMT after SiN passivation, with the inset showing statistical results of V_{th}.

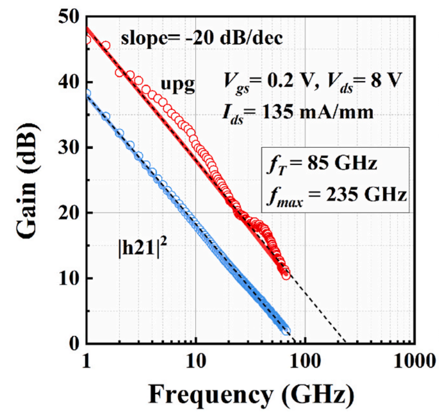
theory [36,37], the minimum access resistance is $R_{ac} = (0.026\,\Omega\,\text{mm})\cdot(10^{13}/n_s)^{0.5}$. The n$^+$-GaN doping concentration (5×10^{19} cm^{-3}) adopted from D-mode devices may be too high for our lower 2DEG density, causing band mismatch that hinders low-bias injection. This non-linearity diminishes at higher V_{gs}. The ultrathin InAlN/AlN polarization layer effectively suppresses short-channel effects in the device, as evidenced by the saturated output current characteristics maintaining a relatively flat profile beyond the knee voltage. The transfer characteristics were measured at a V_{ds} of 2 V, revealing a peak g_m of 492 mS/mm at $V_{gs} = 0.6$ V. A V_{th} of 0.25 V was determined by extrapolating the tangent line at the peak g_m point on the drain current curve. The V_{th} values are reasonably uniform as shown in the inset of Fig. 5(b), with a variation of ~0.08 V. This uniformity primarily attests to the reproducibility and precision of the highly selective, self-terminating gate recess process. No significant thermal shift in the threshold voltage was observed after a post-annealing process at 500 °C. Studies have indicated that in devices fabricated using SF_6-based ICP etching, fluorine incorporation exhibits no measurable impact on the threshold voltage, even following post-annealing at temperatures as high as 800 °C [11,38].

3.3. RF performance and benchmarking

Small-signal characterization from 1 GHz to 67 GHz was performed using a Keysight N5247B network analyzer. An on-wafer SOLT (Short-Open-Load-Thru) calibration was conducted, followed by de-embedding using on-chip open and short structures to eliminate the parasitic pad capacitance and inductance. At a bias of $V_{gs} = 0.2$ V and $V_{ds} = 8$ V, f_T and $f_{\max}$ were extracted to be 85 GHz and 235 GHz, respectively, by extrapolating with a −20 dB/decade slope. The bold solid lines in the figure represent the simulated RF gains based on a small-signal equivalent circuit model, with the extracted parameters listed in the accompanying table; the close agreement between simulation and measurement validates the accuracy of the extracted parameters [2] (Fig. 6). The resulting figure of merit, $f_T \times Lg$, which reflects the electron velocity under the gate and is critical for power amplifier performance, reaches 11.1 GHz μm for the device with Lg = 130 nm. The high $f_T \times$ Lg value can be attributed to the preserved channel mobility owing to the damage-free selective etching, combined with the enhanced 2DEG density from PECVD SiN passivation, which collectively reduced access resistance and improved carrier velocity. As benchmarked against other gate-recessed E-mode GaN HEMTs in the literature (Fig. 7), this value exceeds that of most reported devices with similar gate lengths (100–150 nm) and is competitive with the state-of-the-art. This result stems from the combined benefits of low-damage etching, ultrathin barrier design, and effective passivation, which collectively enhance channel velocity and reduce parasitic resistances. Such a competitive benchmarking position demonstrates the effectiveness of the integrated approach of an ultrathin barrier, a high-selectivity self-terminating recess etch, and an effective passivation in achieving high-frequency performance alongside reliable enhancement-mode operation.

4. Conclusions

In this work, high-performance enhancement-mode GaN HEMTs were achieved through the integration of an ultrathin InAlN/AlN composite barrier, a highly selective self-terminating gate recess etch, and a PECVD SiN passivation layer. The fabricated devices demonstrate a threshold voltage of +0.25 V, a maximum saturation drain current density of 560 mA/mm, and excellent RF performance with $f_T/f_{\max}$ of 85/235 GHz and a competitive $f_T \times$ Lg of 11.1 GHz μm. This integrated approach provides a scalable and low-damage fabrication path for E-mode HEMTs by effectively addressing the dual challenges of precise gate-recess control and access region conductivity recovery. The methodology demonstrates that the combined use of selective etching and interface engineering enables both stable enhancement-mode operation and high-frequency potential. Future work to further advance the performance of such ultrathin-barrier devices may involve optimizing the individual components of this integrated process, particularly the passivation technology (e.g., deposition process, interface control, and thermal stability). These improvements are expected to further enhance device reliability and power density, solidifying the potential of this integrated process for next-generation millimeter-wave power and mixed-signal applications.



g_m (mS/mm)	550
g_d (mS/mm)	25.9
C_{gs} (fF/mm)	615
C_{gd} (fF/mm)	27.3
R_i (Ω·mm)	0.5
R_g (Ω·mm)	0.14
R_s (Ω·mm)	0.36
R_d (Ω·mm)	0.26
$f_{T.model}$ (GHz)	85.1
$f_{max.model}$ (GHz)	228

Model Parameters

Fig. 6. Small-signal RF characteristics and modeled RF parameters for the E-mode HEMT with $L_g = 130$ nm.

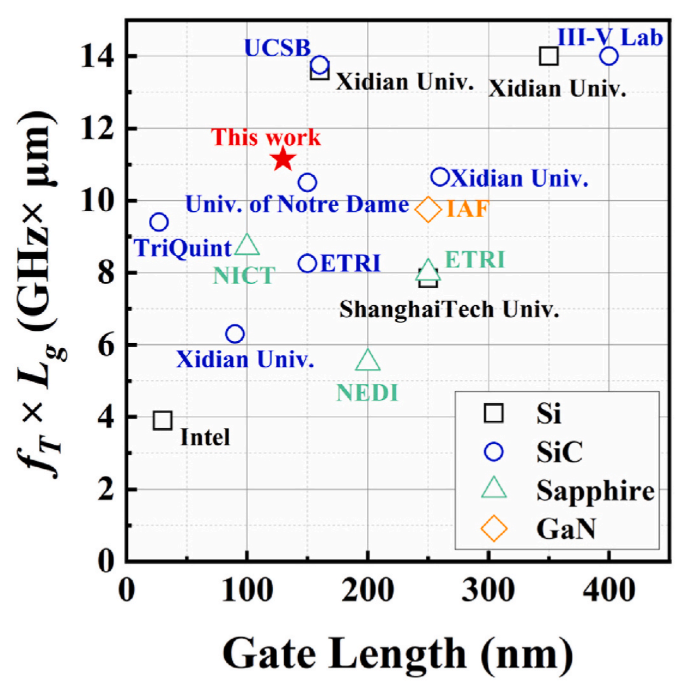


Fig. 7. Benchmark of $f_T \times$ Lg of gate-recessed E-mode GaN HEMTs fabricated on different substrates [15,16,19,39–50].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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